

Doubles for both triple reconstruction and assay validation

torchcell lab notes

Script `experiments/010-kuzmin-tmi/scripts/construction_validation_doubles.py`
Data `experiments/010-kuzmin-tmi/results/construction_validation_doubles.csv`
Dendron `[[experiments.010-kuzmin-tmi.scripts.construction_validation_doubles]]`

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1 Why the pure set-cover was the wrong objective

The essential wet-lab task is a small set of double mutants that both (a) reconstructs many top-ranked triples and (b) has **variance in DMF / interaction** to validate the new echo-plating assay against (versus plate-stamping). The triple-coverage set-cover (`[[experiments.010-kuzmin-tmi.scripts.optimized_doubles_setcover_constructed_10]]`) does (a) with 8 doubles but fails (b): those 8 are all **near-neutral** — Costanzo DMF span 0.15, $|\varepsilon| \leq 0.039$, and **zero significant interactions** — so the assay would have almost nothing to discriminate. An assay validated only on near-1.0 fitness and null interactions has not been shown to detect signal.

2 Two tiers, unioned (13 doubles)

coverage (8) the greedy set-cover doubles; reconstruct **all 31** within-10 top- k triples.

validation (5) added for dynamic range and real signal: the doubles with a **significant Costanzo interaction** ($P < 0.05$ and $|\varepsilon| > 0.08$, both signs), the **highest-DMF** double, and a **low-DMF anchor chosen for quality** — among low-DMF (< 0.75) doubles, the one that also reconstructs triples with the tightest SD (YER079W+YJR060W: DMF 0.610 ± 0.018 , covers 3 triples), *not* the noisy DMF-minimum (YGL087C+YJR060W was 0.513 ± 0.172).

novel (1, separate) **YPL046C+YPL081W (ELC1+RPS9A)**, the one within-10 pair with **no DMF in Costanzo or Kuzmin** — a construction candidate to fill the last cell (Section 6).

Result: 31/31 triples still reconstructed, DMF range **0.610–1.178**, ε from **−0.130** to **+0.098**, **3 significant interactions**.

#	Double	Tier	DMF $\pm$ SD	ε	p	sig	triples
1	YER079W+YJR060W	validation	0.610 ± 0.018	−0.003	0.454		3
2	YER079W+YPL081W	validation	0.863 ± 0.098	−0.130	0.035	✓	0
3	YJR060W+YKL033W-A	validation	0.871 ± 0.023	−0.082	0.007	✓	0
4	YJR060W+YPL046C	coverage	0.966 ± 0.018	+0.003	0.453		4
5	YDR057W+YLL012W	coverage	1.000 ± 0.069	−0.036	0.296		5
6	YLR312C-B+YPL081W	coverage	1.016 ± 0.042	−0.019	0.355		3
7	YLL012W+YPL046C	coverage	1.025 ± 0.053	−0.010	0.429		5
8	YDR057W+YPL081W	coverage	1.035 ± 0.045	+0.039	0.260		3
9	YDR057W+YER079W	coverage	1.096 ± 0.054	+0.012	0.435		5
10	YER079W+YLR312C-B	coverage	1.099 ± 0.039	−0.027	0.290		5
11	YBR203W+YPL046C	coverage	1.118 ± 0.017	+0.036	0.077		5
12	YDR057W+YGL087C	validation	1.137 ± 0.024	+0.098	0.001	✓	3
13	YDR057W+YLR312C-B	validation	1.178 ± 0.026	+0.047	0.119		1
14	YPL046C+YPL081W	novel	—	—	—		0

Fig. 1 The 13 selected doubles by tier, with Costanzo DMF, SD, ε and p .

3 Single mutants to inoculate (10)

To build the doubles above, cross these 10 single-mutant strains pairwise. **Every** panel-10 gene appears in at least one double, so all 10 are required — no smaller set builds them all. The **# doubles** column is how many each gene crosses into (YOS9/YDR057W is the busiest hub at 5; COS111, MMS2 and YKL033W-A each appear once). The novel pair adds no new strain (ELC1 and RPS9A are already in the set).

All 44 measured within-10 doubles ranked by DMF, coloured by tier (coverage, validation, unused; significant ε ringed) — the tiered companion to `constructed_10_dmf_forest`, which predates the validation tier and marks only the 8 set-cover doubles. The 45th pair, YPL046C+YPL081W, is unmeasured and so absent here (Section 6).

DMF, SD, SE, ε and p are the published Costanzo values (SD = sample SD over 4 colonies, directly comparable to the assay’s colony SD — see `[[experiments.010-kuzmin-tmi.scripts.constructed_10_dmf_reference]]`). The low-DMF end is dominated by CBF1/YJR060W (its single defect), so a low DMF there is a fitness effect, not an interaction. Two of the three significant doubles (YER079W+YPL081W, YJR060W+YKL033W-A) enable no top- k triple — they are pure validation targets; the low anchor YER079W+YJR060W and the strongest-positive- ε YDR057W+YGL087C each also rebuild triples. The set is tunable: drop the two triple-less validation doubles for an 11-double set, or add more near-significant ε .

#	Systematic	Common	# doubles
1	YBR203W	COS111	1
2	YDR057W	YOS9	5
3	YER079W	—	4
4	YGL087C	MMS2	1
5	YJR060W	CBF1	3
6	YKL033W-A	—	1
7	YLL012W	YEH1	2
8	YLR312C-B	(SPH1 locus)	3
9	YPL046C	ELC1	4
10	YPL081W	RPS9A	4

Fig. 2 The 10 single-mutant strains to inoculate, with the number of selected doubles each participates in.

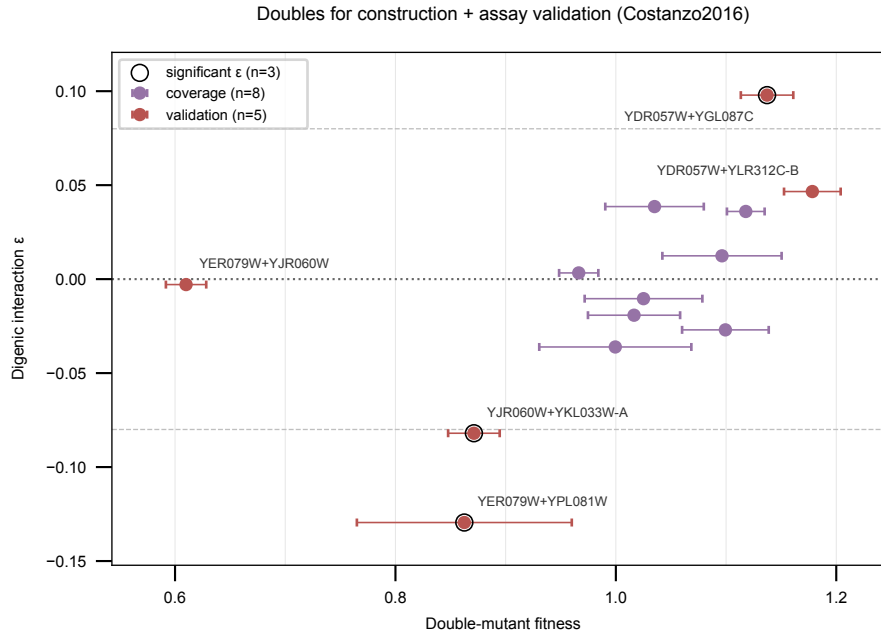


Fig. 3 The selected doubles in interaction-versus-fitness space: Costanzo digenic interaction ε against double-mutant fitness, with the coverage and validation tiers distinguished and the three significant interactions ringed. The validation tier is what spreads the set along both axes; the coverage tier alone clusters near $\varepsilon = 0$.

4 There are only 3 significant interactions in the whole panel

Surveyed across all 66 panel-12 doubles and all three sources: exactly **3** clear Costanzo’s bar ($P < 0.05$ and $|\varepsilon| > 0.08$), and all 3 are within-10 and already in the validation tier — YDR057W+YGL087C (+0.098), YJR060W+YKL033W-A (−0.082), YER079W+YPL081W (−0.130). None hide in the dropped-gene doubles; Kuzmin 2018/2020 have too few measured pairs to reach significance. So “construct the significant ones” means these 3, already selected. The panel is near-neutral by design, so strong interactions are scarce.

4.1 Why a big interaction can still be “not significant” — and why that is fixable

A double’s ε is a point estimate; whether it counts as significant depends on how big it is *relative to its error bar*. The error bar on ε is driven by the colony-to-colony scatter of the double *and* of the two singles it is compared against (all $n = 4$ colonies in Costanzo). So a double sitting on a noisy strain (CBF1, RPS9A) can show a sizable ε that is still inside its own error bar — insignificant. The reverse also happens: YBR203W+YDR057W has a tiny ε (+0.062) that *is* significant ($p = 1.4 \times 10^{-5}$) only because its colonies were very tight (SD 0.010).

The key point for us: **that error bar is set by how many colonies we plate, and that is our choice**. More colonies gives a smaller error bar, shrinking with the square root of the count. So a “noisy but interesting” double is not disqualified — it just needs to be plated deeper. Our goal is depth (measure a few predicted doubles precisely), not breadth (screen many strains shallowly), which is exactly what echo dispensing is good at. So we pick the validation doubles for *interesting values* (fitness extremes, large ε) and drive each to a tight measurement at the bench.

All 44 measured within-10 doubles by DMF
color = tier; ring = significant ϵ

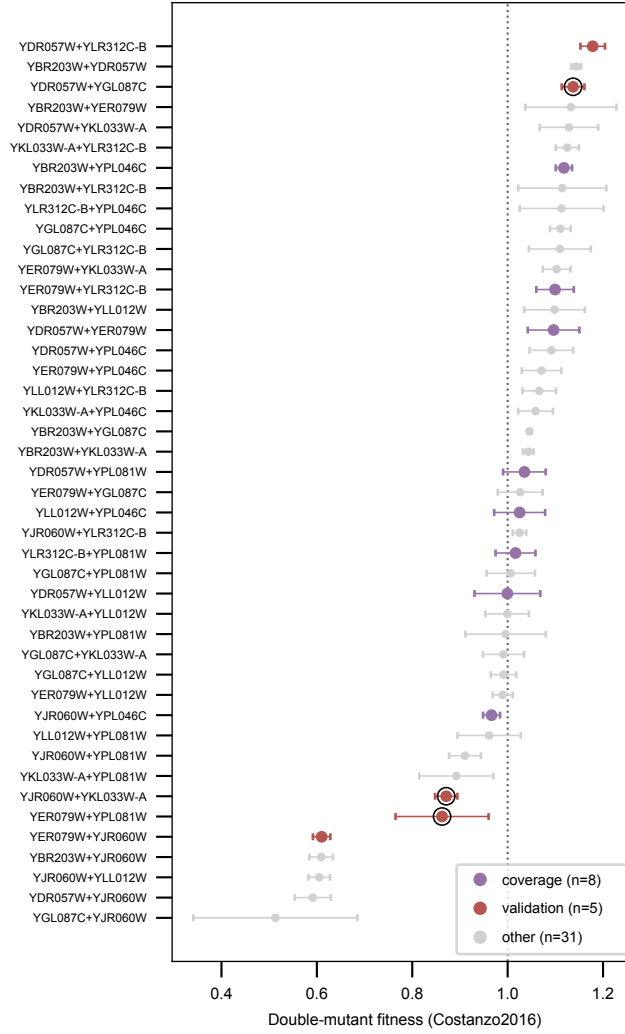


Fig. 4 Forest plot of all 44 measured within-10 doubles, ranked by Costanzo DMF and coloured by tier.

One caveat: deeper plating tightens *our* error bar, but Costanzo’s stays at $n = 4$. Once ours is tighter, a disagreement may be the reference’s noise, not ours.

5 Cross-dataset check — Costanzo versus Kuzmin DMF (a real caveat)

The DMF values above are **Costanzo only**. Where Kuzmin also measured a within-10 double, the two references sometimes **disagree sharply**, especially for the low-DMF CBF1 doubles:

double	Costanzo DMF	Kuzmin DMF
YJR060W+YLL012W	0.605	1.119 (K2020)
YJR060W+YPL046C	0.966	0.738 (K2018)
YBR203W+YJR060W	0.609	0.768 (K2018)
YGL087C+YPL046C	1.111	0.998 (K2018)

CBF1/YJR060W’s low DMF is largely a *Costanzo* phenomenon — Kuzmin reads those pairs higher, and YJR060W+YLL012W disagrees by about 0.5. So the low-DMF “signal” our assay would validate against is reference-dependent; treat the low-DMF anchor’s target loosely and, where a Kuzmin value exists, report both. The full per-double Costanzo+Kuzmin comparison is in the supplementary table (Section 7).

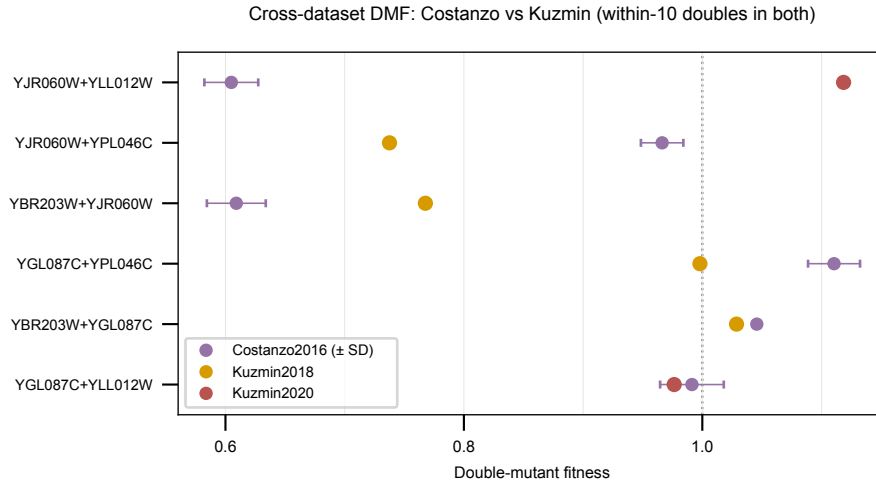


Fig. 5 Dumbbell of every within-10 double measured in both references: Costanzo with SD, joined to the Kuzmin value.

5.1 The same ambiguity exists *within* Costanzo, at the strain level

The disagreement above is between *datasets*. There is a second, easier-to-miss ambiguity *inside* Costanzo: a gene can carry more than one strain, with materially different measurements. YJR060W (CBF1) is the case that matters for this panel —

Costanzo strain	SMF	stddev
YJR060W_dma2646	0.9230	0.0102
YJR060W_sn154	0.5900	0.1138

— two published single-mutant values 0.33 fitness units apart. The queried reference table the assay compares against picked **sn154** by a dictionary collision in `12_panel_inference_3_queried_data_tables.py` (the gene-keyed index is overwritten by each successive record, so the surviving row is whichever LMDb iteration reached last), not by any stated rule. The same code applies no temperature filter, so 26 °C versus 30 °C is also last-record-wins — moot for this panel, where all 12 ORFs happen to carry identical 26/30 entries, but not moot in general (17% of Costanzo strains differ between the two temperatures).

This matters because CBF1 anchors the low end of every fitness correlation we report. It needs a deliberate strain-selection rule before those correlations carry weight.

6 Novel construction candidate — the one unmeasured pair

Of $\binom{10}{2} = 45$ pairs, **44 have a Costanzo DMF; 1 does not: YPL046C+YPL081W (ELC1+RPS9A)** — and it is also absent from Kuzmin 2018/2020. Missing is not the same as synthetic-lethal: an SL pair would show as a measured low fitness or an explicit flag, not a blank. A NaN just means the pair was never in a scored query×array orientation.

SL check (SynthLethDB Yeast_SL.csv). **not synthetic-lethal.** ELC1/YPL046C appears in **zero** SL records, and RPS9A/YPL081W is SL only with its paralog **RPS9B** — the classic redundant-duplicate case — not with ELC1.

BioGRID / SGD second source. Checked via SGD’s aggregated interaction API: ELC1 has **97** recorded genetic-interaction partners and **RPS9A is not among them** — no interaction of any type (SL, negative or positive genetic, synthetic growth defect). ELC1 is well studied at 97 partners, so RPS9A’s absence is a real gap, not sparse coverage.

Recommendation. Construct it. It is the one cell of the 10×10 matrix with no value in Costanzo, Kuzmin, SynthLethDB or BioGRID, and it is not synthetic-lethal, so it should be buildable. Tagged **novel** in the supplementary table.

7 Supplementary table — all 45 within-10 doubles

`experiments/010-kuzmin-tmi/results/construction_validation_doubles.csv` is the full supplementary table: **every** within-10 pair, with `tier` blank when not selected (`coverage / validation / novel` otherwise), Costanzo DMF

$\pm$ SD, derived SE, ε , p , the **Kuzmin 2018/2020 DMF $\pm$ SD** columns, and the strain id. 45 rows = $\binom{10}{2}$; kept as a CSV rather than inlined because its roughly 15 columns are too wide to render at this page width.

8 What was actually built

Of the 14 doubles this selection called for, **13 were constructed**. **YKL033W-A $\times$ YJR060W (CBF1) failed to construct** (reported by Aurosh Sharma on handover of the 2026-08-06 plating). It is absent from the run-4 strain list, so the built panel is 12 singles + 13 doubles + WT. The missing pair is a construction failure, not a dropout to impute.

The plating that resulted is recorded in `experiments/W019-echo-crispr-array/data/run4_doubles_2026-08-06/PROVENANCE.md`: three independently randomized plates, WT at 28 wells and each mutant at 14 wells per plate, with both constituent singles of every double present on the same plate so $\varepsilon = f_{ab} - f_a f_b$ is computable within-plate.

9 Run 4 — the constructed panel, measured

The doubles selected above were built and plated on 2026-08-06 as three independently randomized 384-well plates (WT at 28 wells, every mutant at 14, 378 transfers + 6 blanks per plate; full provenance in `experiments/W019-echo-crispr-array/data/run4_doubles_2026-08-06/PROVENANCE.md`). Scoring is by `experiments/W019-echo-crispr-array/scripts/run4_doubles_48h.py`.

No triple mutants. This round carries **singles and doubles only** — the strain list has a K01 and a K02 column and 26 rows (WT, s1–s12, d1–d13). There is structurally no third-knockout field, so nothing here bears on trigenic interaction; that remains ahead of us.

Registration. All three plates resolved orientation confidently on strain structure and returned **6/6 blank wells empty**. That last number is load-bearing: this collaborator’s Echo export *omits* the blank wells rather than listing them, so the blanks are reconstructed as the complement of the 378 transfers. Six of six empty means the reconstruction and the orientation are both right, checked against the image rather than assumed.

plate	occupied	blanks empty	WT CV	QC
P1	320/384	6/6	0.153	pass
P2	345/384	6/6	0.158	pass
P3	369/384	6/6	0.122	pass

9.1 Single-mutant fitness reproduces

Against the published Costanzo SMF the singles come back at **Pearson $r = 0.706$** ($p = 0.015$, $n = 11$). Eleven, not twelve, because this panel’s s7 is **YLR104W (LCL2)** where run 3 carried YPL056C (LCL1) — the two rounds’ single panels differ by one strain, and the join is on the systematic ORF rather than on position in the list.

9.2 Singles and doubles on one axis

Both measurements come from the same three plates and the same code, so they belong on the same axis — and that is where the asymmetry shows. The singles track the identity line; the doubles sit low and flat, compressed into a narrow band regardless of what the reference says they should be.

9.3 Double-mutant fitness does not

The same comparison for the 13 constructed doubles gives $r = 0.255$ ($p = 0.42$, $n = 12$) — twelve because YPL046C+YPL081W, the novel pair, has no published DMF anywhere, which is why it was selected. Singles reproducing while doubles do not is the first sign of the problem the next section is about, and it is not a coincidence: the two are measured on the same plates by the same code.

10 Digenic interaction versus Costanzo — measured, and not yet usable

The design supports the comparison cleanly. Both constituent singles of every double sit on the same plate, so

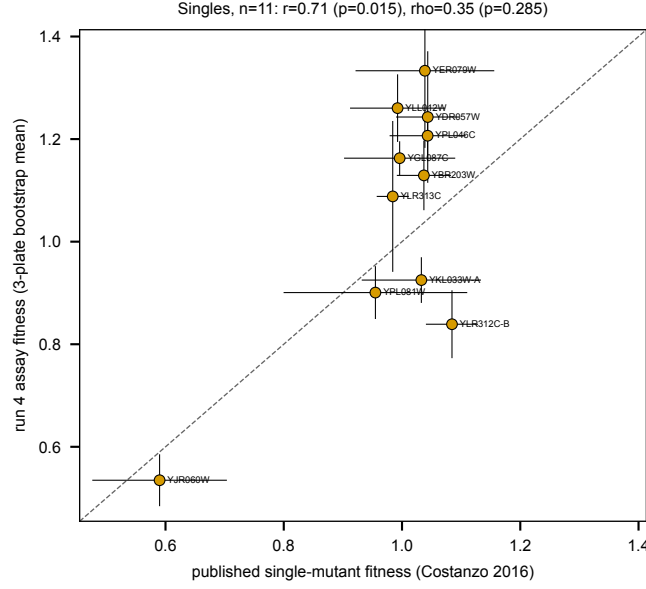


Fig. 6 Run 4 single-mutant fitness (bootstrap mean over the three re-randomized plates, plate as the resampling unit) against published Costanzo 2016 SMF. Identity dashed; error bars are SE-vs-SE.

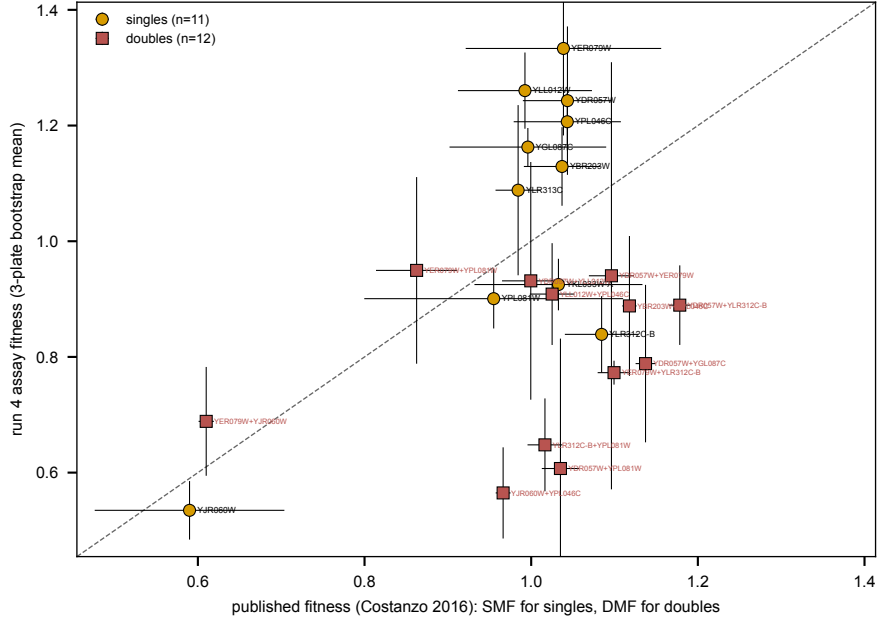


Fig. 7 Run 4 against published Costanzo values on one axis: single mutants (circles, SMF) and double mutants (squares, DMF). Identity dashed. The singles spread along the line; the doubles do not track it.

$$\varepsilon_{ab} = f_{ab} - f_a f_b \quad (1)$$

is formed entirely within a plate — no cross-plate normalization enters the interaction term, and the three plates give three independent replicates of the same quantity, which is what the across-plate bootstrap turns into an error bar.

Run against the published Costanzo ε for the same pairs, the answer is $r = -0.245$ ($p = 0.44$, $n = 12$): no agreement, and the sign is wrong.

10.1 Every double, side by side

The full comparison, one row per constructed double, sorted by the published ε . The table is generated by `experiments/W019-echo-crispr-array/scripts/run4_doubles_48h.py` straight from `run4_doubles_vs_reference.csv` — no number here is transcribed by hand.

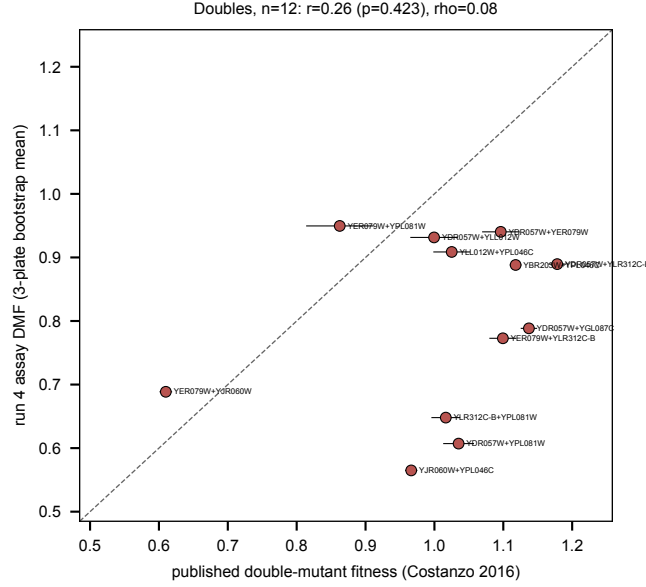


Fig. 8 Run 4 double-mutant fitness against published Costanzo DMF. The 13th double, the novel YPL046C+YPL081W, is absent because no published value exists for it.

double	f_{ab}	$f_a f_b$	ours				Costanzo	
			ε	SE _{within}	SE _{across}	p	ε	sig.
YER079W+YPL081W	0.950	1.224	-0.274	0.046	0.195	0.295	-0.130	✓
YDR057W+YLL012W	0.932	1.570	-0.639	0.057	0.253	0.127	-0.036	
YER079W+YLR312C-B	0.773	1.092	-0.319	0.061	0.025	0.006	-0.027	
YLR312C-B+YPL081W	0.648	0.747	-0.099	0.046	0.100	0.426	-0.019	
YLL012W+YPL046C	0.909	1.519	-0.611	0.062	0.108	0.030	-0.010	
YER079W+YJR060W	0.689	0.733	-0.045	0.089	0.117	0.739	-0.003	
YJR060W+YPL046C	0.565	0.653	-0.088	0.077	0.097	0.462	+0.003	
YDR057W+YER079W	0.940	1.712	-0.772	0.054	0.454	0.231	+0.012	
YBR203W+YPL046C	0.888	1.372	-0.483	0.063	0.149	0.083	+0.036	
YDR057W+YPL081W	0.607	1.139	-0.532	0.047	0.275	0.193	+0.039	
YDR057W+YLR312C-B	0.890	1.023	-0.133	0.058	0.085	0.257	+0.047	
YDR057W+YGL087C	0.788	1.442	-0.654	0.056	0.170	0.061	+0.098	✓
YPL046C+YPL081W	0.934	1.094	-0.161	0.043	0.131	0.344	—	

Read the $f_a f_b$ column first. It is the multiplicative expectation, and for ten of the thirteen pairs it exceeds 1.0 — which cannot be right for two deletion mutants and is what drives every ε in the fourth column negative. YPL046C+YPL081W has no published ε because it is the novel pair; it was constructed precisely because no reference value exists.

10.2 Our own significance — and which error term you use decides everything

Section 4.1 argued that a big-but-insignificant interaction is fixable because the error bar is set by how many colonies we plate, and that is our choice. This round is the first test of that argument, and the answer is more interesting than expected: **colony count was never the binding constraint**.

Two error terms can be put on ε , and the table reports both.

Within-plate. ε is a function of three strain means measured on the same plate, so by the delta method

$$\text{var}(\varepsilon) = \text{var}(f_{ab}) + f_b^2 \text{var}(f_a) + f_a^2 \text{var}(f_b), \quad (2)$$

each variance being $(\text{colony SD}/\sqrt{n})^2$. This is the term Costanzo’s own p -value is built from, and the one that 14 colonies per plate rather than 4 was supposed to shrink. It averages **0.058** across the thirteen doubles.

Across-plate. The scatter of the three per-plate ε values, which carries the batch effect the within-plate term cannot see. It is **larger than the within-plate term for 12 of 13 doubles**, by a median factor of **2.4** and up to **8.5**.

That gap is the finding. Judged on colony scatter alone, **11 of 13** doubles would clear $p < 0.05$ (median $|\varepsilon|/\text{SE} = 5.9$). Judged honestly, on three independent plate-level estimates, **2 of 13** do. Plating deeper does buy precision within a plate — it just buys precision on a quantity whose plate-to-plate reproducibility is the actual limit. The lever for this assay is *more plates*, not more colonies per plate, which is the opposite of what Section 4.1 assumed.

Read the p column with the caveat below. The test is a two-sided t -test on three plate-level estimates, $df = 2$ — weak by construction, so a non-significant call here is poor evidence of absence. And more importantly: with the WT shift of Section 10.4 unresolved, this p measures how *reproducibly* we recover a biased ε , not whether that ε is right. Precision is not accuracy. The two doubles that do clear $p < 0.05$ (YER079W+YLR312C-B, YLL012W+YPL046C) are *not* the two Costanzo calls significant, and neither should be reported as a detected interaction.

10.3 Why this is an artifact and not a result

It should not be read as a failure to reproduce Costanzo’s interactions. Four observations, all from experiments/W019-echo-crispr-array/results/run4_interactions.csv, identify it as a normalization problem upstream of Equation (1):

diagnostic	value
singles scoring <i>above</i> the on-plate WT	8 of 12 (up to 1.333)
multiplicative expectation $f_a f_b > 1$	10 of 13 pairs (up to 1.712)
sign of ε	negative for all 13
$\text{corr}(\varepsilon, f_a f_b)$	−0.926 ($p = 5 \times 10^{-6}$)

A multiplicative null above 1.0 for two deletion mutants is not a meaningful expectation, and an ε that correlates at −0.93 with its own expectation term is tracking the singles, not the doubles. Concretely, for the strongest published interaction in the panel:

YDR057W+YGL087C	f_a	f_b
run 4 fitness	1.243	1.163
expectation $f_a f_b$		1.446
observed f_{ab}		0.788
ε (ours)		−0.654
ε (Costanzo)		+0.098

That −0.654 is arithmetic from two inflated singles, not a measured epistasis.

10.4 Where the inflation comes from

The singles are on the same scale as run 3’s — just shifted. The eleven strains common to both rounds correlate at $r = 0.685$, but run 4 sits a near-uniform $1.26\times$ above run 3 (median ratio; dividing it out leaves a mean absolute difference of 0.111). In run 3 every single scored *below* WT (0.67–0.92); in run 4 most score above it.

This matters for ε and not for the correlations because ε is **not scale-invariant**. Under a uniform inflation $f \rightarrow kf$,

$$\varepsilon' = kf_{ab} - k^2 f_a f_b, \quad (3)$$

so the expectation term grows quadratically while the observed term grows linearly, driving ε negative in proportion to $f_a f_b$ — exactly the −0.926 correlation observed. A Pearson correlation, by contrast, is invariant to k , which is why the singles still reproduce at $r = 0.706$ while every interaction is corrupted.

10.5 What is not yet known

Why the WT reference moved is untested. At the colony level the WT median is *higher* than the mutant median on all three plates (ratio 0.90–0.94), so this is not simply sick or contaminated WT wells. Run 3 named its wild type BY4741 explicitly; run 4’s strain list gives it as ‘-’. Whether that is a different stock, a different plating density, or something else has not been checked, and no correction should be applied before it is — rescaling to make the numbers look right would bury the question.

The cheapest next diagnostic is to re-score run 4 against the *plate median* instead of the WT wells. If the singles fall back onto run 3’s scale, the problem is localized to the WT wells specifically and ε becomes computable; if they do not, it is something broader in the round and belongs at the bench.

11 Related notes

- [[experiments.010-kuzmin-tmi.scripts.topk_triples_from_constructed_10]]

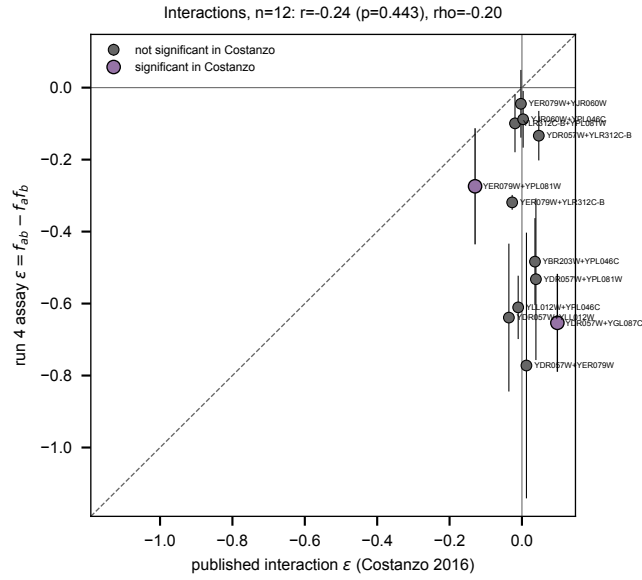


Fig. 9 Run 4 ε against published Costanzo ε , the two Costanzo-significant pairs marked. Every point sits below zero and the published spread is not recovered. This panel is published as the record of a diagnosed artifact, not as a measurement of epistasis.

- `[[experiments.010-kuzmin-tmi.scripts.optimized_doubles_setcover_constructed_10]]`
- `[[experiments.010-kuzmin-tmi.scripts.constructed_10_dmf_reference]]`
- `[[experiments.010-kuzmin-tmi.12_panel_crispr_fitness_assay]]` — the assay these doubles validate